



Midterm Exam Fall 2025

YOUR NAME: _____

YOUR SEMINAR INSTRUCTOR'S NAME: _____

--- This is a double-sided exam ---

Welcome to the Midterm Exam for Frontiers of Science. Please **write your name and your seminar instructor's name on this exam**. The exam begins at **2:00 pm** and finishes at **4:00 pm**, giving you **2 hours** to complete it.

This exam consists of four questions: two Mind and Brain questions (15 points each; parts A, B & C) and two Physics and Reality questions (15 points each; parts A, B & C). You must answer ALL questions in the exam packet within the empty space that follows each question. Should you need extra space, you may use the empty pages in the back of the exam. Feel free to use your calculator as necessary. To receive full credit, show all your work and include units where applicable. Be sure to turn in this exam packet!

As a guideline for your answers, each question is marked with: [NC] – 'requires no calculations', [SC] – 'requires simple calculations or quantitatively reading a graph', or [C] – 'requires calculations'. **Good luck!**

Numerical Constants, Values, and Unit Conversions

c (Speed of light)	$3.00 \times 10^8 \text{ m s}^{-1}$
h (Plank's constant)	$6.63 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$
G (Gravitational constant)	$6.67 \times 10^{-11} \text{ m}^3 \text{ s}^{-2} \text{ kg}^{-1}$
R_E (Radius of Earth)	$6.38 \times 10^6 \text{ m}$
M_E (Mass of Earth)	$5.97 \times 10^{24} \text{ kg}$
M_S (Mass of Sun)	$1.99 \times 10^{30} \text{ kg}$
r_p (Radius of proton)	$8.33 \times 10^{-16} \text{ m}$
r_H (Radius of hydrogen atom)	$5.29 \times 10^{-11} \text{ m}$
m_p (Mass of proton)	$1.67 \times 10^{-27} \text{ kg}$
m_e (Mass of electron)	$9.11 \times 10^{-31} \text{ kg}$

Metric System Prefixes

T	"tera"	10^{12}
G	"giga"	10^9
M	"mega"	10^6
k	"kilo"	10^3
c	"centi"	10^{-2}
m	"milli"	10^{-3}
μ	"micro"	10^{-6}
n	"nano"	10^{-9}
p	"pico"	10^{-12}

$$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 6.24 \times 10^{18} \text{ eV}$$

$$1 \text{ mile} = 1.61 \text{ km}$$

$$1 \text{ foot} = 0.305 \text{ m}$$

$$1 \text{ parsec (pc)} = 3.26 \text{ light years (ly)} = 3.09 \times 10^{16} \text{ m}$$

$$1 \text{ metric ton} = 1000 \text{ kg}$$

$$1 \text{ lb} = 0.454 \text{ kg}$$

$$1 \text{ L} = 10^{-3} \text{ m}^3$$

$$1 \text{ gallon} = 3.79 \text{ L}$$

$$\pi = 3.14$$

Distance

$$d = v t$$

d = distance, v = velocity, t = time

Gamma factor

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

γ = gamma factor, v = velocity (speed), c = speed of light

Time dilation and length contraction

$$(\text{time elapsed for moving object}) = \frac{(\text{time elapsed for stationary object/observer})}{\gamma}$$

$$(\text{length of moving object}) = \frac{(\text{length of stationary object})}{\gamma}$$

Energy of a massive particle

$$E = \gamma m_r c^2$$

m_r = rest mass of object, c = speed of light

Schwarzschild radius

$$r_s = \frac{2GM}{c^2}$$

r_s = Schwarzschild radius, G = gravitational constant, M = mass of object, c = speed of light

Gravitational time dilation

$$T = t \times \left(1 - \frac{2GM}{rc^2}\right)^{1/2}$$

T = time at a distance r from a massive object's center, t = time in empty space, G = gravitational constant, M = mass of object, c = speed of light, r = distance from a massive object's center

De Broglie Wavelength

$$\lambda = \frac{h}{mv}$$

λ = wavelength, h = Planck's constant, m = mass of object, v = speed

Heisenberg Uncertainty Principle

$$\Delta x \Delta v \geq \frac{h}{4\pi m}$$

Δx = uncertainty in position, Δv = uncertainty in velocity (speed), h = Planck's constant, m = mass of object

Geometry

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3$$

$$V_{\text{cylinder}} = h \pi r^2$$

V = volume, r = radius

V = volume, r = radius, h = height

Density

$$D = \frac{M}{V}$$

D = density, M = mass of object, V = volume

Statistics

$$SE = \frac{SD}{\sqrt{N}}$$

SE = standard error of the mean, SD = standard deviation,

N = sample size

MIND & BRAIN

Question 1. [15 points]

Scientific studies suggest that active learning is important for memory. Researchers hypothesize that the way individuals engage with their environment influences how accurately they recall different types of environmental details, such as landmarks, spatial order, and the relationships among environmental elements.

To test this hypothesis, Plancher and colleagues designed three experimental scenarios in a virtual reality setting, where participants navigate a town from inside a car. There were a total of 72 participants (36 females and 36 males) recruited from the undergraduate psychology department. One third of participants (12 females and 12 males) were assigned to each group. The three groups were: (1) Passive group, where each participant was a passenger in a car in a fixed itinerary; (2) Planning-only group, where each participant chose the itinerary, but was only a passenger in the car; (3) Interaction-only group, where each participant drove the car in a fixed itinerary.

1A. [5 points, SC]

One aspect of memory researchers tested was how well participants remembered specific landmarks in the city (Environmental Elements), regardless of their relation to the itinerary. Their results are shown in Figure 1 below.

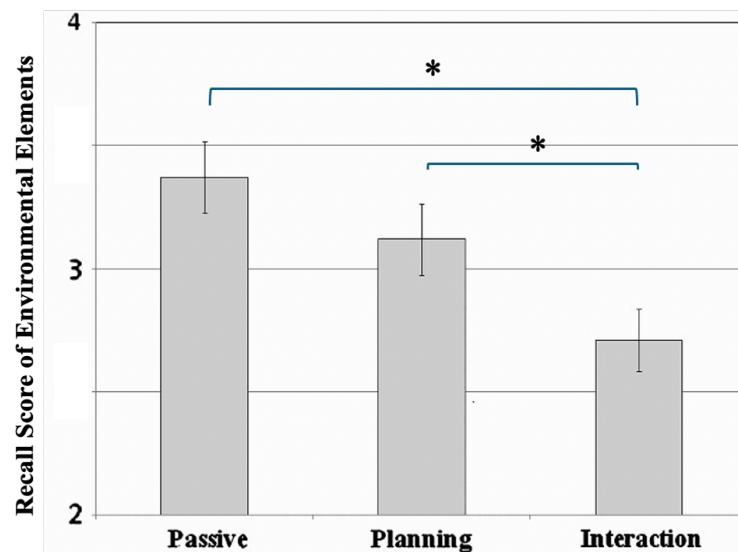


Figure 1. Recall score of environmental elements for each group, where 4 corresponds to 100% recall and 0 corresponds to 0% recall. Asterisk (*) indicates $p < .05$.

1A. (i) Is there a difference in the recall score of environmental elements between the three groups? Explain using your knowledge of statistics.

1A. (ii) Which brain region do you think is involved in the recognition of environmental elements that participants saw while virtually navigating the city? One word answer, please.

1A. (iii) Researchers also tested how well participants could orient themselves and remember the itinerary-related elements of the route (e.g. the desired location was northwest of this corner, or 10 blocks up and 2 blocks left, etc.). Their results are shown in Figure 2 below.

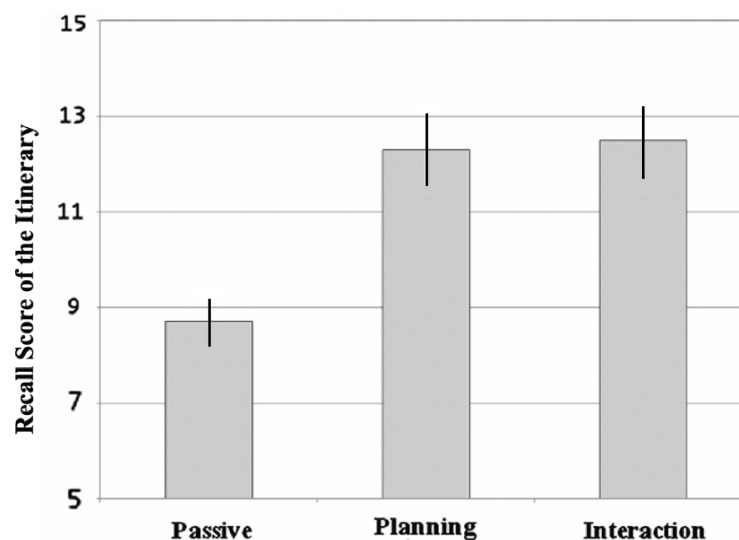


Figure 2. Recall score of the itinerary for each group, where 15 corresponds to 100% recall and 0 corresponds to 0% recall. Error bars correspond to $\pm 1SE$.

1B. (i) Is there a difference in the recall score of the itinerary between the three groups? Explain using your knowledge of statistics.

1B. [5 points, SC]

1B. (i) Based on the data presented in Figures 1 and 2, is the researcher's hypothesis supported? Explain in 1-2 sentences.

1B. (ii) Based on the data presented in Figures 1 and 2, name the group that was most successful in their overall recall and in 1-2 sentences, propose an explanation for why that might be so.

1B. (iii) Imagine the following scenario: you and your two best friends, Ella and Maya, are visiting Rio for the first time. You are in charge of planning the trip itinerary, Ella is in charge of navigating with a map, and Maya is in charge of taking Insta-worthy pictures as you explore the city.

Based on what you learned from Plancher's study, speculate who might best recognize the following details of your trip? Enter Y for you, E for Ella and M for Maya:

Christ the Redeemer, The Sugarloaf Rock, The Lapa Aqueduct, the tile pattern along the beach promenade. _____

The street names and necessary turns taking them through the north-south sequence of beaches: Leme, Copacabana, Ipanema, and Leblon. _____

1C. [5 points, NC]

1C. (i) In 1-2 sentences, explain why this study does not use a typical control group.

1C. (ii) Name two limitations of the study. Explain each limitation in one sentence.

1C. (iii) In 2-3 sentences, propose your own follow-up study that would provide converging evidence to Plancher and colleague's study findings.

Question 2. [15 points]

MRI machines are classified by the strength of a magnetic field they can create, measured in the units of tesla (T). Colizoli and colleagues (2021) compared the results obtained by a 3 T machine to that of a 7 T machine, trying to determine which machine works best for their research. The task used in this study required participants to be able to identify whether dots in an image are moving up or down, at two levels of difficulty. At the *Easy* level, 85% of the dots move in the same direction (either up or down) while 15% move in random directions. At the *Hard* level, 70% of the dots move in the same direction, while 30% move in random directions.

2A. [5 points, C]

2A. (i) An fMRI machine with a stronger magnetic field allows for more detailed imaging. However, researchers are concerned that the stronger magnetic field could affect the participants' performance (Percent Correct) on the visual tasks. **Based on the results presented in Table 1 below, would you say the concerns of the researchers are justified?** Explain using your knowledge of statistics.

Table 1. *Performance of eight human participants in the moving dots visual task at two difficulty levels and in two fMRI machines.*

Difficulty	3 T Percent Correct	Standard Deviation		7 T Percent Correct	Standard Deviation
Hard	71.3	3.9		67.2	4.1
Easy	89.5	3.1		86.9	3.4

2A. (ii) Based on Table 1, did the difficulty level of the task affect participants' performance on the task? Explain using your knowledge of statistics.

2A. (iii) One of your friends is looking for a research paper about Reward Prediction Error and habit formation to use as a reference for their semester project. **Considering the task used by Colizoli and colleagues and the type of decision participants had to make, would you recommend your friend read their paper?** Explain your reasoning.

2B. [5 points, SC]

2B. (i) You are assisting another group of researchers interested in investigating the neural processes of decision making. Specifically, they are interested in the differences in brain activity between hard and easy decisions. Figure 3 on the next page shows the results of such a comparison done by Colizoli and colleagues. **How would you interpret a zero value along the y-axis in Figure 3?**

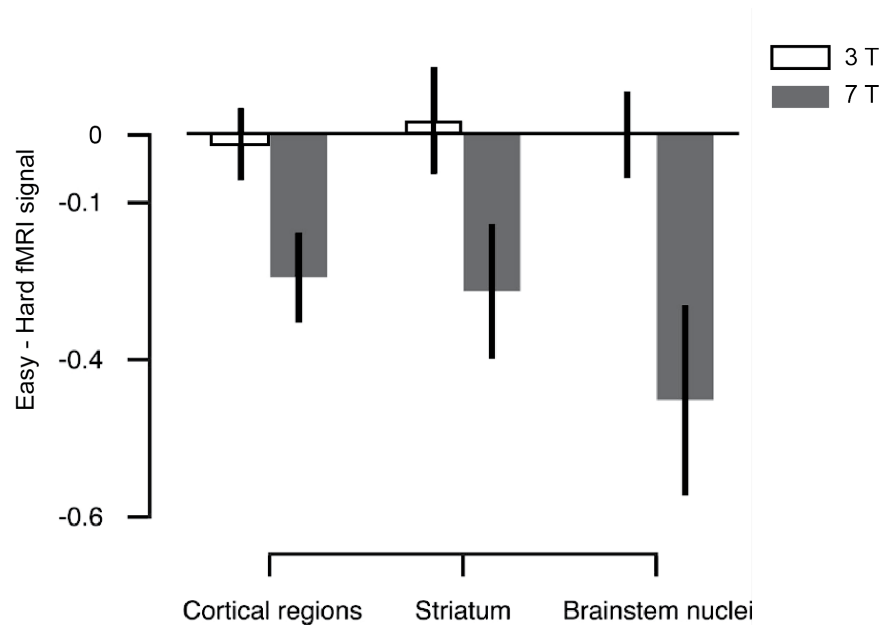


Figure 3. Difference in fMRI signal between two difficulty levels in each of the three brain regions. The y-axis shows the difference of the fMRI signal while participants are working on the Hard visual task compared to the Easy visual task (Easy - Hard). The data are shown for different brain regions (cortical, striatum, and brainstem nuclei). Error bars represent ± 1 SE.

2B. (ii) Based on the results presented in Figure 3, which machine works better for investigating the effect of the difficulty of decisions related to visual tasks, as described in 2B. (i)? Explain using your knowledge of statistics.

2B. (iii) Considering the study published by Colizoli and colleagues (2021), classify each potential error below as either *systematic* (S) or *random* (R).

- () The number of coordinated moving dots is at 85% for all trials, even when they were supposed to be Hard.
- () The 7 T machine is not operating at full power, generating a magnetic field of 3 T.
- () Participants come to the study with a range of energy levels and familiarity with these types of visual tasks.
- () Some of the participants are trained in meditation techniques and can lay in the fMRI machine perfectly still for many hours, while other participants easily feel bored and have a hard time laying still for more than a few minutes
- () Participants are not given earplugs to protect them from the loud noises from the fMRI machine, making the task more difficult for them.

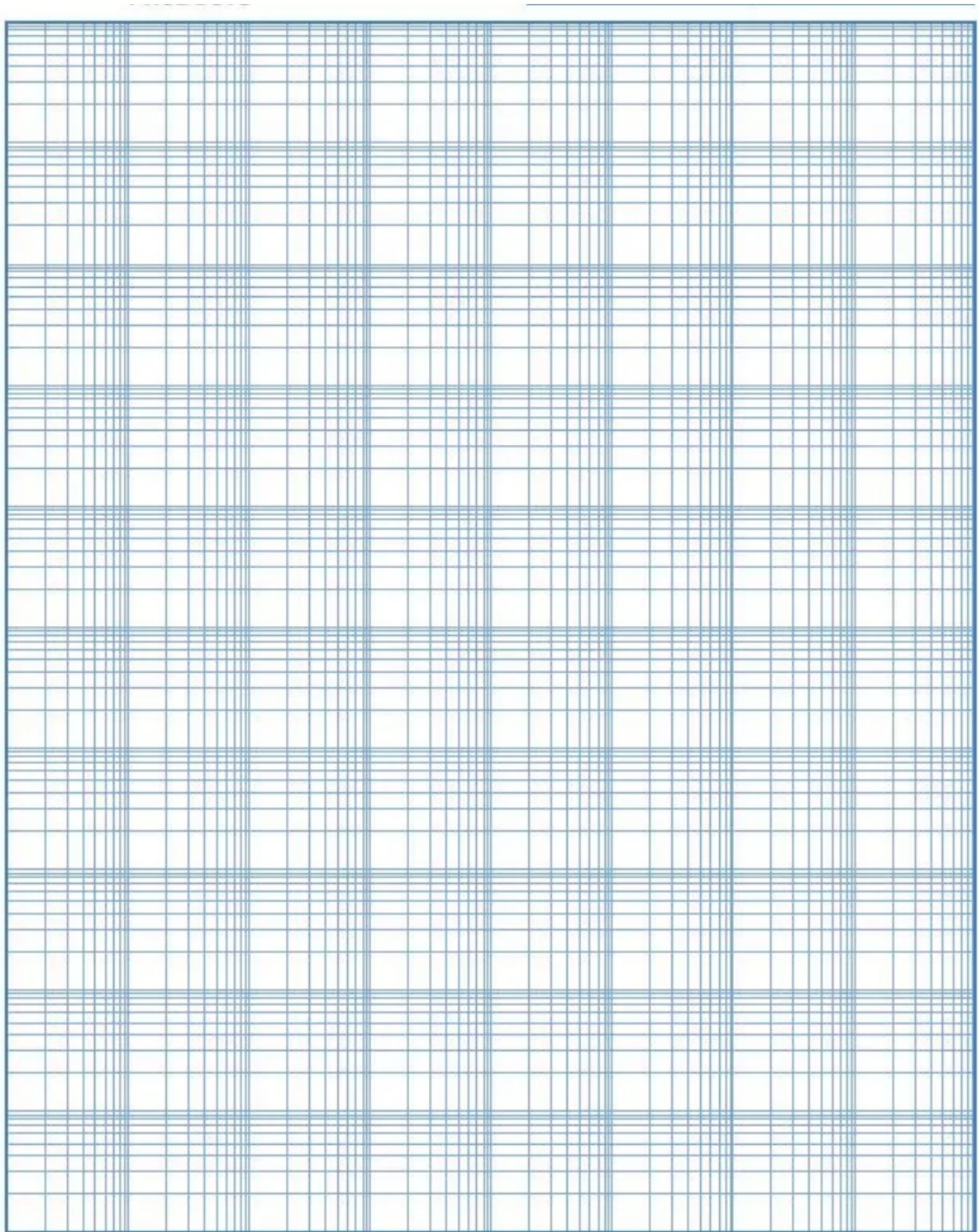
2C. [5 points, C]

Let's consider the scales involved in the brains of different organisms. Table 2 below lists the brain mass and total number of neurons in the brain of several animals:

Table 2. *Brain mass and total neuron count for different animals.*

<i>Species</i>	<i>Brain Mass (grams)</i>	<i>Total Neuron Count</i>
Mouse	0.4	2.08×10^7
Beagle	80	2.40×10^9
Rhesus Monkey	90	6.39×10^9
Human	1,400	8.54×10^{10}
Elephant	5,000	2.55×10^{11}

2C. (i) Plot brain mass versus total neuron count on the graph in the next page. Label the axes accordingly and introduce an appropriate scale.



2C. (ii) The total volume of all the neurons in the human brain is $8.60 \times 10^{-5} \text{ m}^3$. Approximate each neuron as a cube with a $10 \text{ }\mu\text{m}$ side. **If we arrange all those neurons side by side, one neuron thick, what surface area would these neurons cover?**

PHYSICS & REALITY

Question 3. [15 points]

Alice is piloting the Voyager spaceship, traveling past Earth at $0.8c$. A diagram of her voyage is shown in Figure 4 below. Along the way, mission control on Earth sets up some tests to explore relativity.

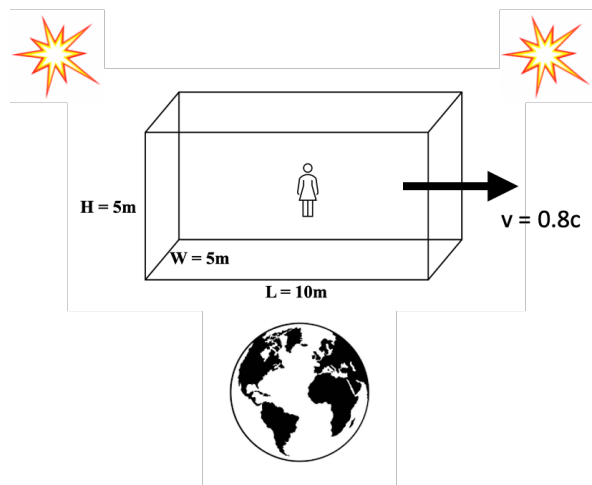


Figure 4. The dimensions of Alice's ship and her velocity (indicated by the arrow) from Earth's perspective. The bursting stars represent two supernovae explosions. Figure not to scale.

3A. [5 points, C]

3A. (i) Just when Alice is passing by Earth, mission control on Earth observes two supernovae explode simultaneously, equidistant from Alice. The position of the supernovae is shown in Figure 4 above. **According to Alice, which of the following is correct?**

- a. The supernovae explode simultaneously.
- b. The supernova at the front of her ship explodes first.
- c. The supernova at the back of her ship explodes first.
- d. The explosions are simultaneous for all observers because light travels at c .

3A. (ii) According to Alice, what can she say about the dimensions of her spaceship? Enter the signs $=$, $>$, or $<$ to indicate equal to, greater than, or less than in the blanks below:

L ____ 10 m

W ____ 5 m

H ____ 5 m

3A. (iii) While on her journey, Alice carefully monitors her health indicators. In particular, she measures that her heart rate is 70 beats per minute. **What is Alice's heart rate according to mission control on Earth?**

3B. [5 points, C]

Voyager parks at a distance 36 billion meters away from the center of a black hole she wants to study. Alice observes that when 1 hour passes on her spaceship, 2.5 hours pass for Bob, her colleague who is far away in empty space.

3B. (i) Given this time difference, how many Schwarzschild radii away is Alice from the center of the black hole?

3B. (ii) Calculate the Schwarzschild radius of the black hole in units of m.

3B. (iii) Calculate the mass of the black hole in units of kg.

3C. [5 points, NC]

Fascinated with her calculations, Alice begins to further contemplate the nature of Life, the Universe, and Everything.

3C. (i) She considers Bob, who is now in a sealed box with no windows. Inside the box there are only two items: an apple with a mass of 100 g, and a digital weight scale (of standard construction, designed to work in Earth's gravity). Bob places the apple on the weight scale, which reads 100 g. **Which of the following scenarios, as illustrated in Figure 5 below, could Bob be in?**

- a. A & B
- b. B & D
- c. C & D
- d. B, C, and D

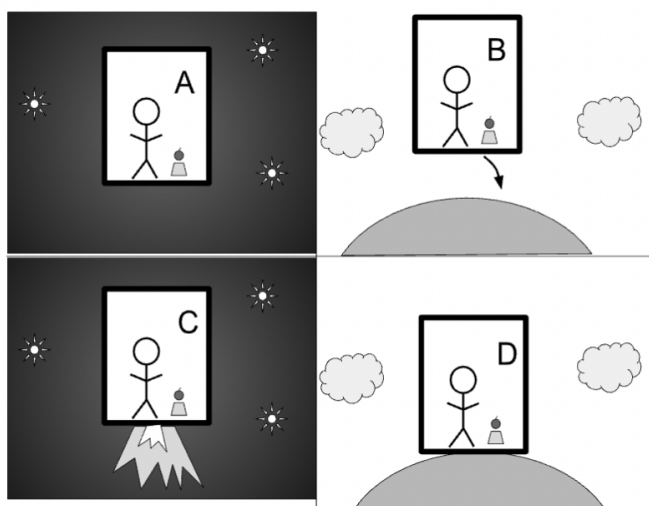


Figure 5. Bob, his apple, and his scale in four different scenarios.

3C. (ii) Lastly, Alice notices that light bends when passing near a massive object, as if it is being deflected. **Explain, in 1-2 sentences, why gravity bends light even though photons have no mass.**

Question 4. [15 points]

In 1988, more than 60 years after de Broglie's proposal of the wave-like properties of matter, scientists performed the double-slit experiment that was described in Professor Greene's lecture using neutrons. Neutrons are microscopic particles with a mass of 1.67×10^{-27} kg.

4A. [5 points, SC]

Two different experimental setups were used during the experiment: neutrons were sent through an impenetrable wall that had either a single-slit or a double-slit opening. The results for each setup are shown in Figure 6 below.

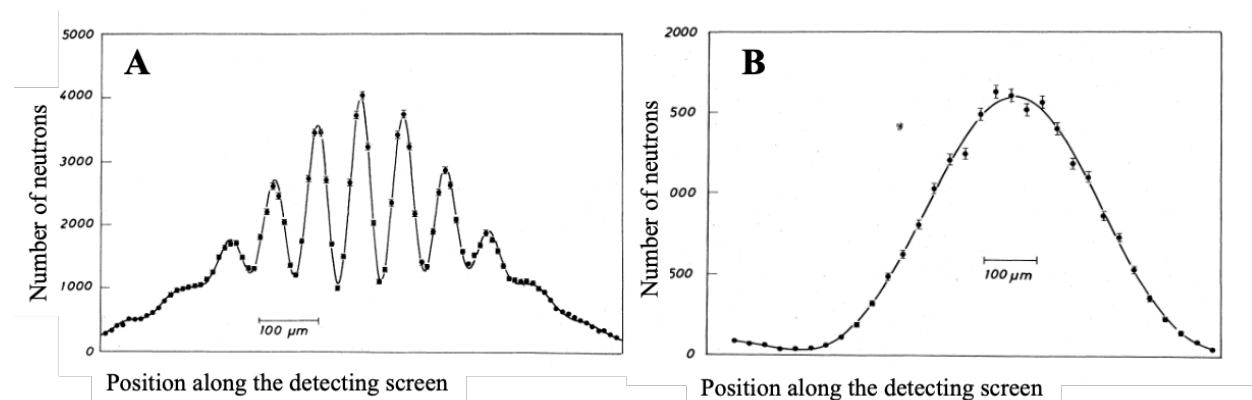


Figure 6. The results of the single- and double-slit setups. The solid curves represent theoretical prediction, and the dots represent the experimental measurements.

4A. (i) Consider plots A and B shown in Figure 6 above. **Which one corresponds to the single-slit and which one corresponds to the double-slit setup?** Explain each choice in one sentence.

4A. (ii) Re-examine the plot that according to you corresponds to the double-slit setup. **Is there any feature of this plot that does not exactly match what you would expect from what you learned about the behavior of electrons in a double-slit experiment?**

4A. (iii) Suppose the experimenters start to increase the distance between the slits. **Draw below or describe in words the kind of pattern you would eventually expect once the distance between the slits becomes very large.**

4B. [5 points, C]

The researchers report that the wavelength of the neutrons used in the experiment was $\lambda = 1.845 \pm 0.14$ nm. The uncertainty in wavelength corresponds to a 7.6% uncertainty in the velocity of the neutrons.

4B. (i) Consider the de Broglie equation for λ . **What was the uncertainty in the velocity of the neutrons?** Express your answer in units of m/s.

4B. (ii) **What was the minimum uncertainty in the position of the neutrons?** Express your answer in units of nm.

4C. [5 points, C]

4C. (i) What is the rest energy of a neutron? Express your answer in units of J.

4C. (ii) Given your calculations in part 4B, do you expect the energy of the neutrons used in the experiment to be much different from what you calculated in 4C. (i)? Hint: you do not need to perform a calculation here.